

Task1 Linux Shell学习报告

姓名：邹墨 邮箱：mozhibubai@qq.com

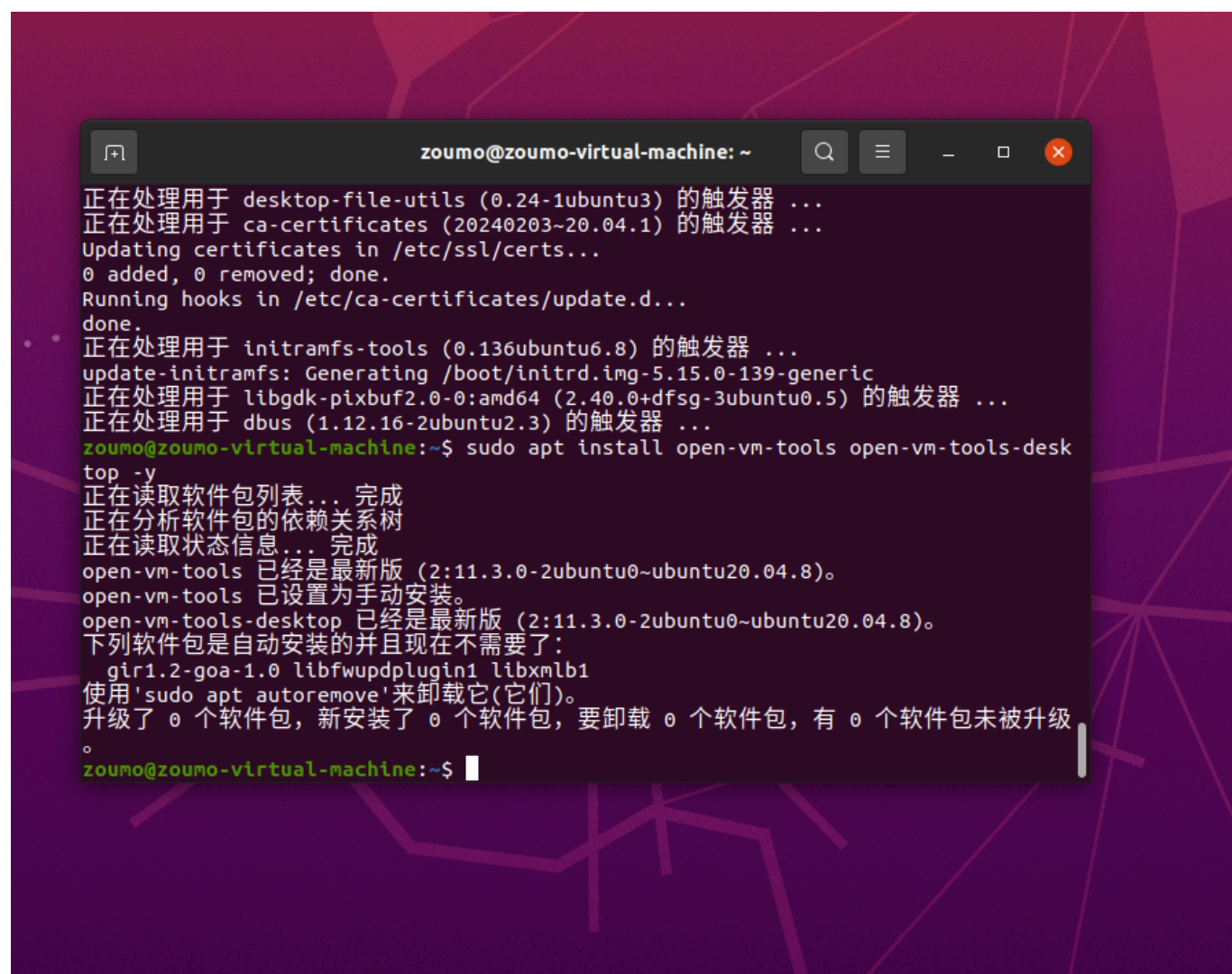
一、学习目标

熟悉Linux基础Shell命令，掌握文件操作、目录切换、管道、重定向等基础操作，理解命令行工作方式，完成实验练习。

二、命令学习与实验截图

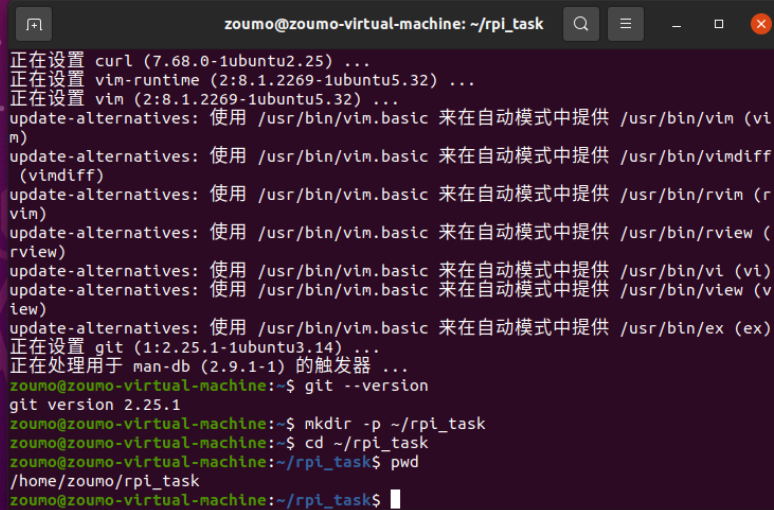
2.1 ls 命令

```
ls
```



2.2 cd 命令

```
cd shell_demo
```

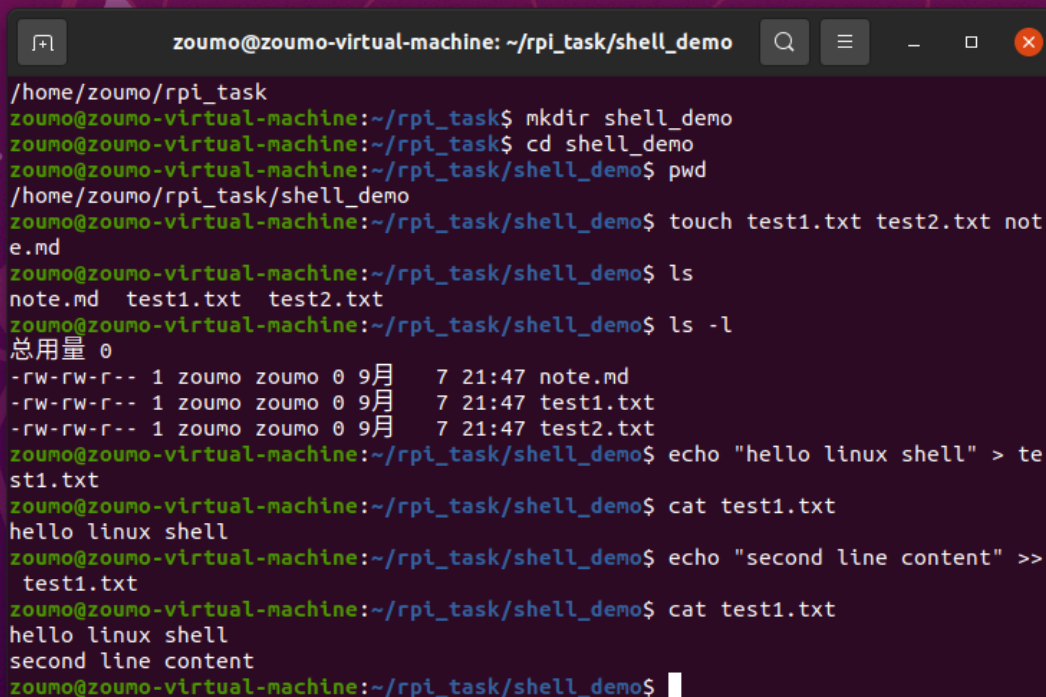


A terminal window titled 'zoumo@zoumo-virtual-machine: ~/rpi_task'. The terminal shows the following commands and output:

```
正在设置 curl (7.68.0-1ubuntu2.25) ...
正在设置 vim-runtime (2:8.1.2269-1ubuntu5.32) ...
正在设置 vim (2:8.1.2269-1ubuntu5.32) ...
update-alternatives: 使用 /usr/bin/vim.basic 来在自动模式中提供 /usr/bin/vim (vim)
update-alternatives: 使用 /usr/bin/vim.basic 来在自动模式中提供 /usr/bin/vimdiff (vimdiff)
update-alternatives: 使用 /usr/bin/vim.basic 来在自动模式中提供 /usr/bin/rvim (rvim)
update-alternatives: 使用 /usr/bin/vim.basic 来在自动模式中提供 /usr/bin/rview (rview)
update-alternatives: 使用 /usr/bin/vim.basic 来在自动模式中提供 /usr/bin/vi (vi)
update-alternatives: 使用 /usr/bin/vim.basic 来在自动模式中提供 /usr/bin/view (view)
update-alternatives: 使用 /usr/bin/vim.basic 来在自动模式中提供 /usr/bin/ex (ex)
正在设置 git (1:2.25.1-1ubuntu3.14) ...
正在处理用于 man-db (2.9.1-1) 的触发器 ...
zoumo@zoumo-virtual-machine:~$ git --version
git version 2.25.1
zoumo@zoumo-virtual-machine:~$ mkdir -p ~/rpi_task
zoumo@zoumo-virtual-machine:~$ cd ~/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ pwd
/home/zoumo/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$
```

2.3 文件查看 cat 命令

```
cat test.txt
```

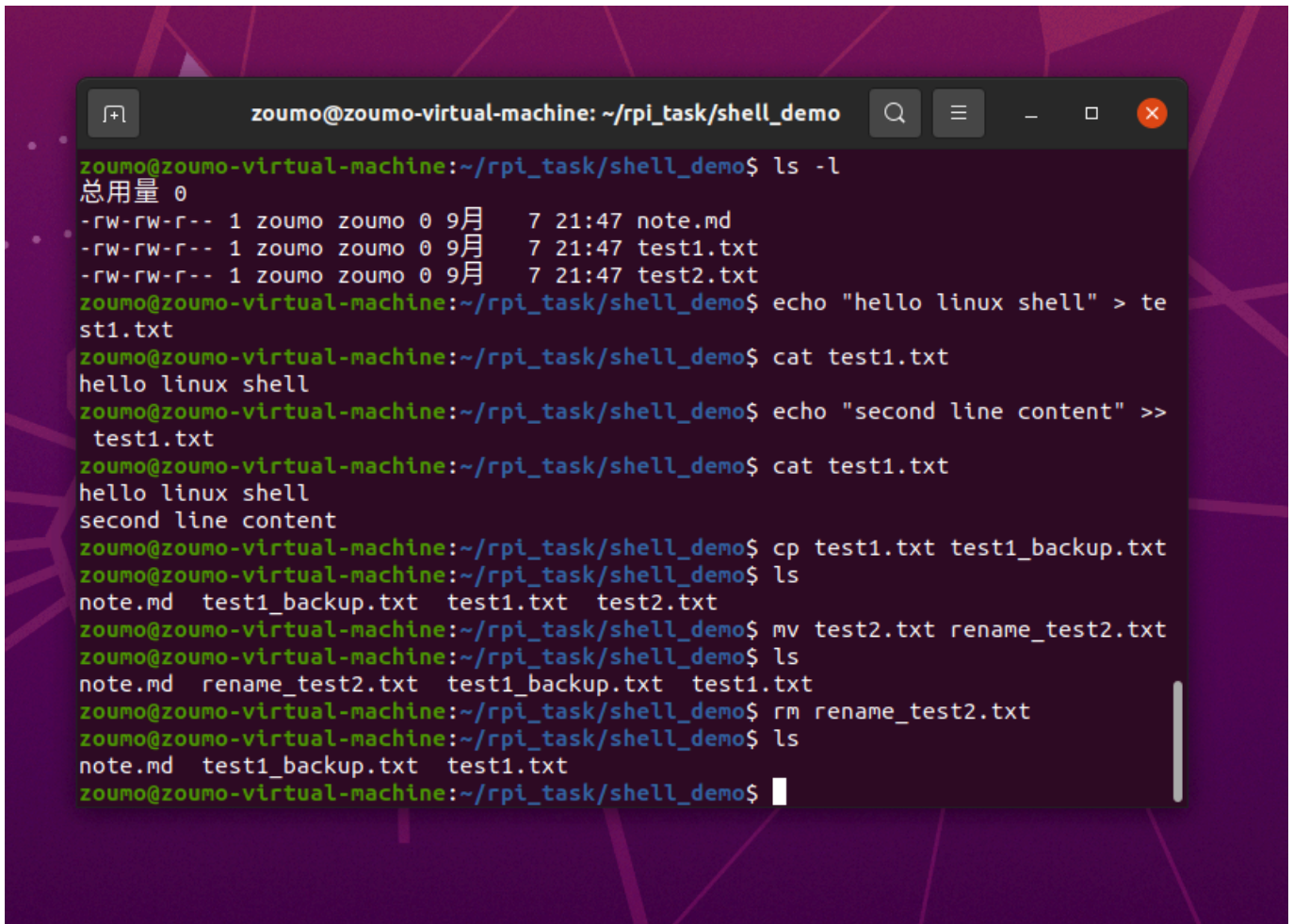


A terminal window titled 'zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo'. The terminal shows the following commands and output:

```
/home/zoumo/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ mkdir shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task$ cd shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ touch test1.txt test2.txt note.md
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md test1.txt test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls -l
总用量 0
-rw-rw-r-- 1 zoumo zoumo 0 9月 7 21:47 note.md
-rw-rw-r-- 1 zoumo zoumo 0 9月 7 21:47 test1.txt
-rw-rw-r-- 1 zoumo zoumo 0 9月 7 21:47 test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo "hello linux shell" > test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1.txt
hello linux shell
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo "second line content" >> test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1.txt
hello linux shell
second line content
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.4 echo 输出与写入文件

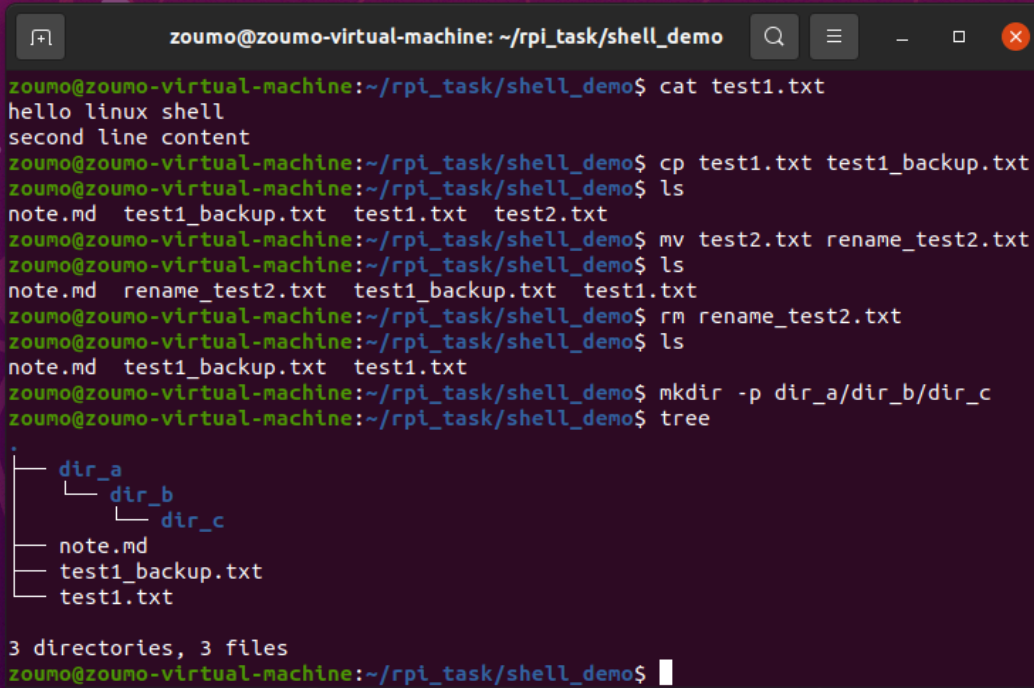
```
echo "hello" > test.txt
```

A terminal window titled 'zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo' with standard window controls. The terminal shows a series of commands and their outputs. It starts with 'ls -l' showing files 'note.md', 'test1.txt', and 'test2.txt'. Then 'echo "hello linux shell" > test1.txt' is executed. Next, 'cat test1.txt' shows 'hello linux shell'. Then 'echo "second line content" >> test1.txt' is executed. Then 'cat test1.txt' shows both lines. Then 'cp test1.txt test1_backup.txt' is executed. Then 'ls' shows the updated file list including 'test1_backup.txt'. Then 'mv test2.txt rename_test2.txt' is executed. Then 'ls' shows 'rename_test2.txt' instead of 'test2.txt'. Finally, 'rm rename_test2.txt' is executed, and the final 'ls' shows the files after removal.

```
zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls -l
总用量 0
-rw-rw-r-- 1 zoumo zoumo 0 9月 7 21:47 note.md
-rw-rw-r-- 1 zoumo zoumo 0 9月 7 21:47 test1.txt
-rw-rw-r-- 1 zoumo zoumo 0 9月 7 21:47 test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo "hello linux shell" > test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1.txt
hello linux shell
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo "second line content" >> test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1.txt
hello linux shell
second line content
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cp test1.txt test1_backup.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md test1_backup.txt test1.txt test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ mv test2.txt rename_test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md rename_test2.txt test1_backup.txt test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ rm rename_test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md test1_backup.txt test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.5 cp 文件复制命令

```
cp test.txt test2.txt
```

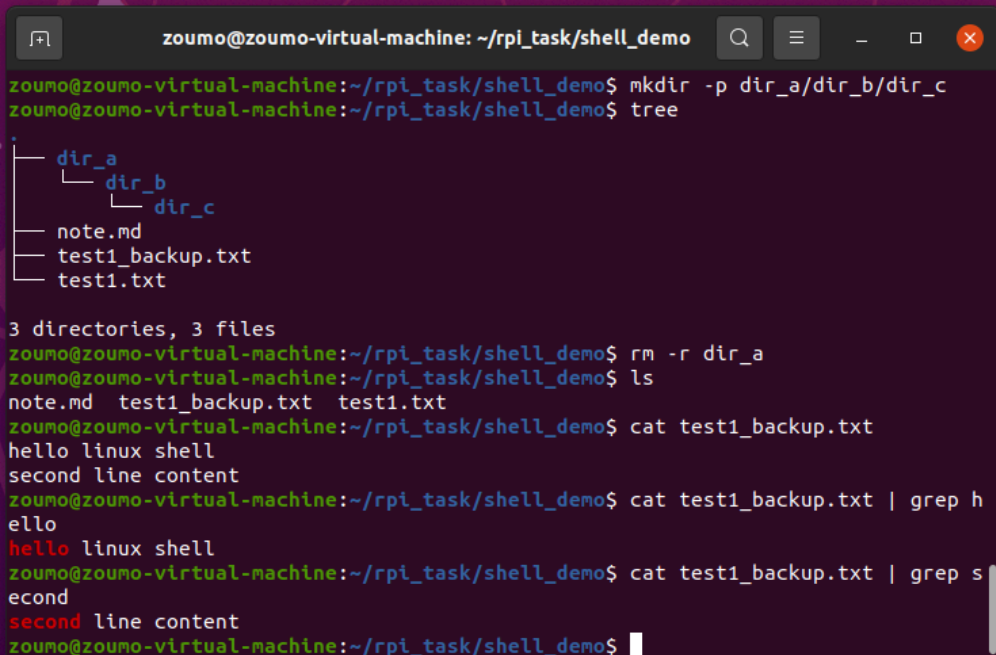


```
zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1.txt
hello linux shell
second line content
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cp test1.txt test1_backup.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md test1_backup.txt test1.txt test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ mv test2.txt rename_test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md rename_test2.txt test1_backup.txt test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ rm rename_test2.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md test1_backup.txt test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ mkdir -p dir_a/dir_b/dir_c
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ tree
.
├── dir_a
│   └── dir_b
│       └── dir_c
├── note.md
├── test1_backup.txt
└── test1.txt

3 directories, 3 files
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.6 mv 文件移动 / 重命名

```
mv test2.txt test_new.txt
```

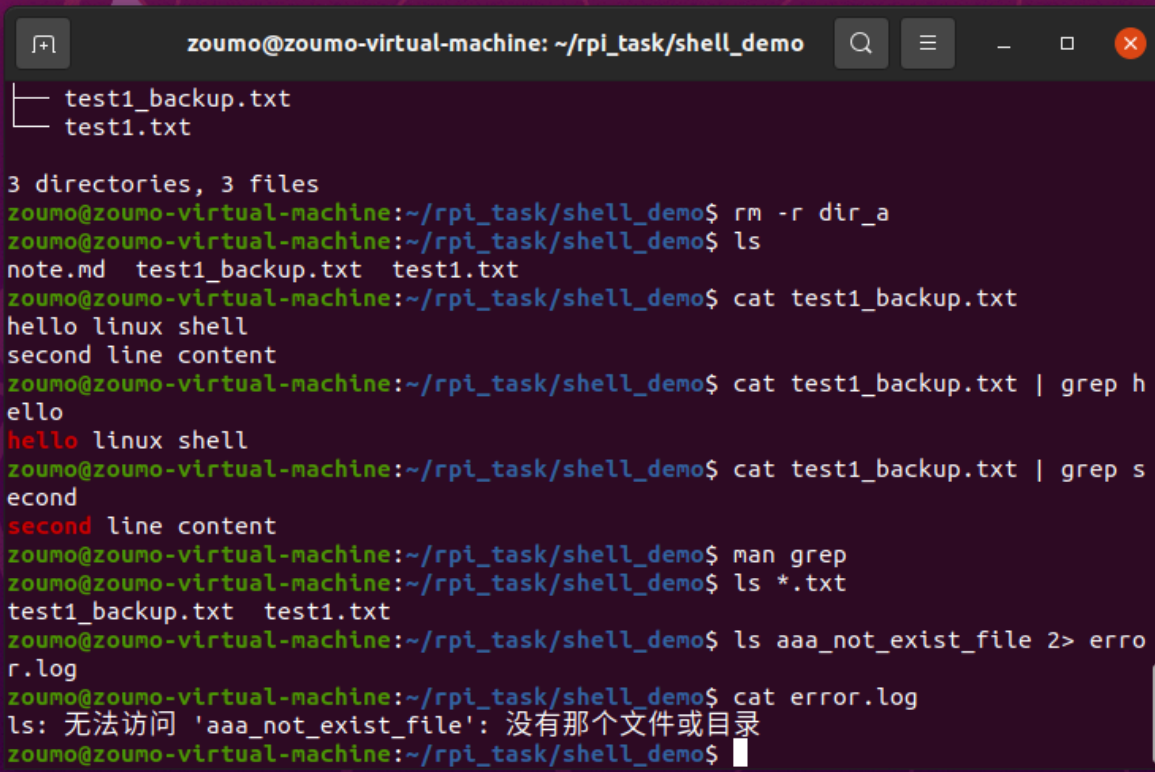


```
zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ mkdir -p dir_a/dir_b/dir_c
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ tree
.
├── dir_a
│   └── dir_b
│       └── dir_c
├── note.md
├── test1_backup.txt
└── test1.txt

3 directories, 3 files
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ rm -r dir_a
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md test1_backup.txt test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1_backup.txt
hello linux shell
second line content
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1_backup.txt | grep h
ello
hello linux shell
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1_backup.txt | grep s
econd
second line content
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.7 rm 删除文件

```
rm test_new.txt
```



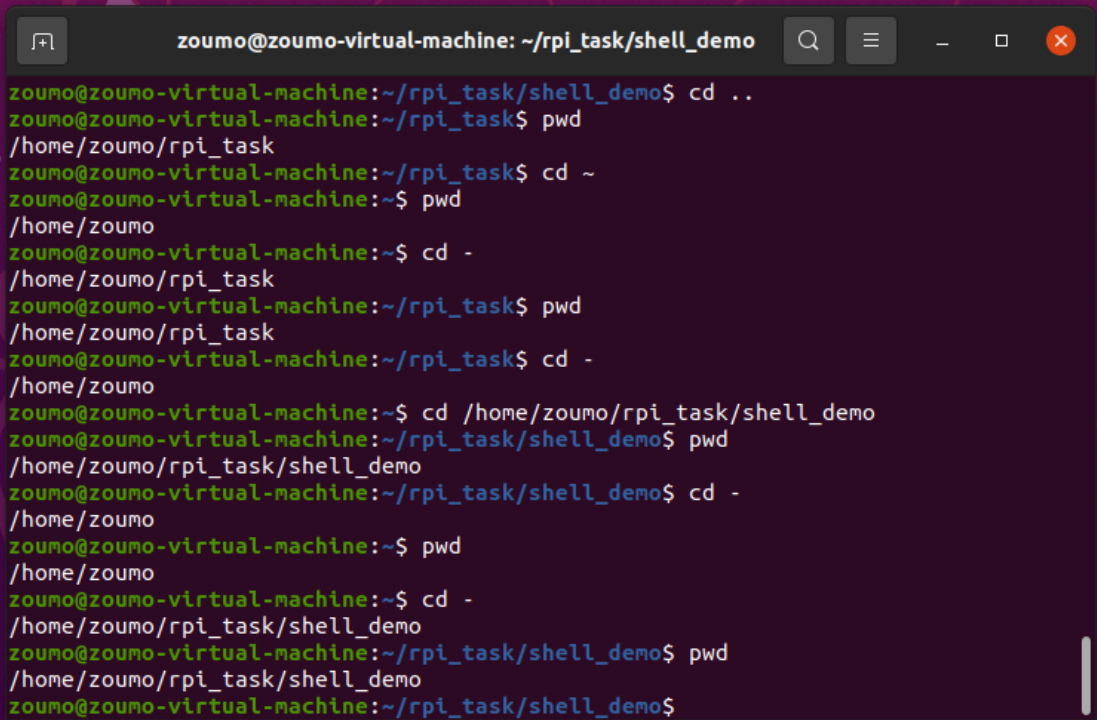
A terminal window titled 'zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo' showing a series of commands and their outputs. The terminal has a dark background with a file explorer icon on the left. The commands and outputs are as follows:

```
└─ test1_backup.txt
└─ test1.txt

3 directories, 3 files
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ rm -r dir_a
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls
note.md test1_backup.txt test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1_backup.txt
hello linux shell
second line content
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1_backup.txt | grep h
ello
hello linux shell
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat test1_backup.txt | grep s
econd
second line content
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ man grep
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls *.txt
test1_backup.txt test1.txt
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ ls aaa_not_exist_file 2> erro
r.log
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cat error.log
ls: 无法访问 'aaa_not_exist_file': 没有那个文件或目录
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.8 管道 | 和 grep 筛选

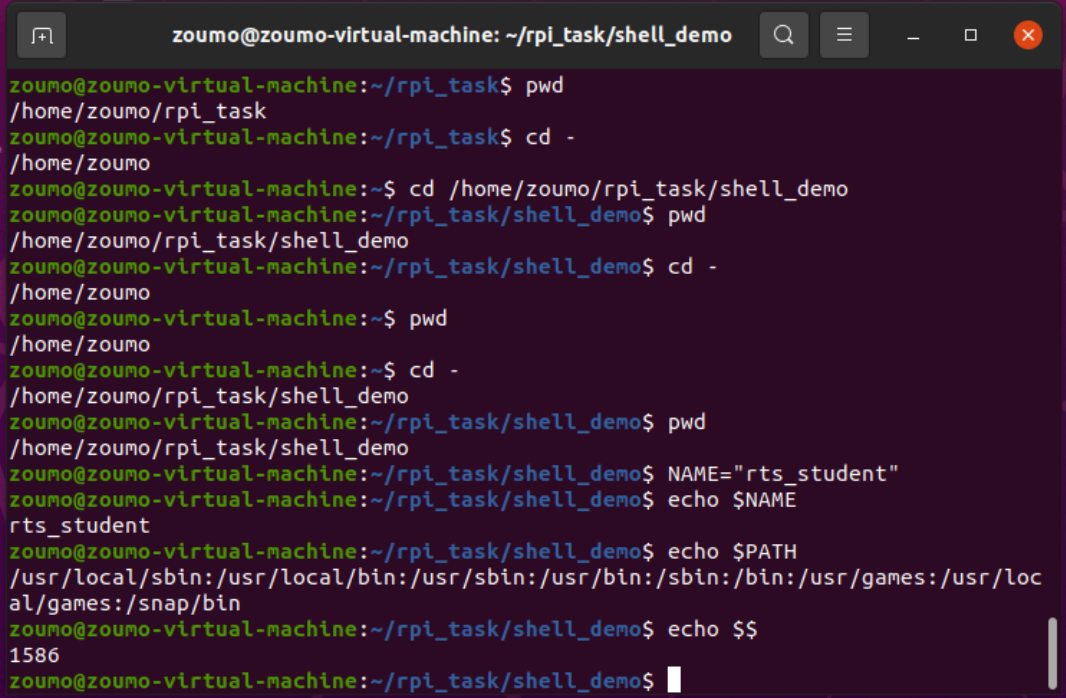
```
ls | grep demo
```



```
zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cd ..
zoumo@zoumo-virtual-machine:~/rpi_task$ pwd
/home/zoumo/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ cd ~
zoumo@zoumo-virtual-machine:~$ pwd
/home/zoumo
zoumo@zoumo-virtual-machine:~$ cd -
/home/zoumo/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ pwd
/home/zoumo/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ cd -
/home/zoumo
zoumo@zoumo-virtual-machine:~$ cd /home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cd -
/home/zoumo
zoumo@zoumo-virtual-machine:~$ pwd
/home/zoumo
zoumo@zoumo-virtual-machine:~$ cd -
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.9 通配符 * 使用

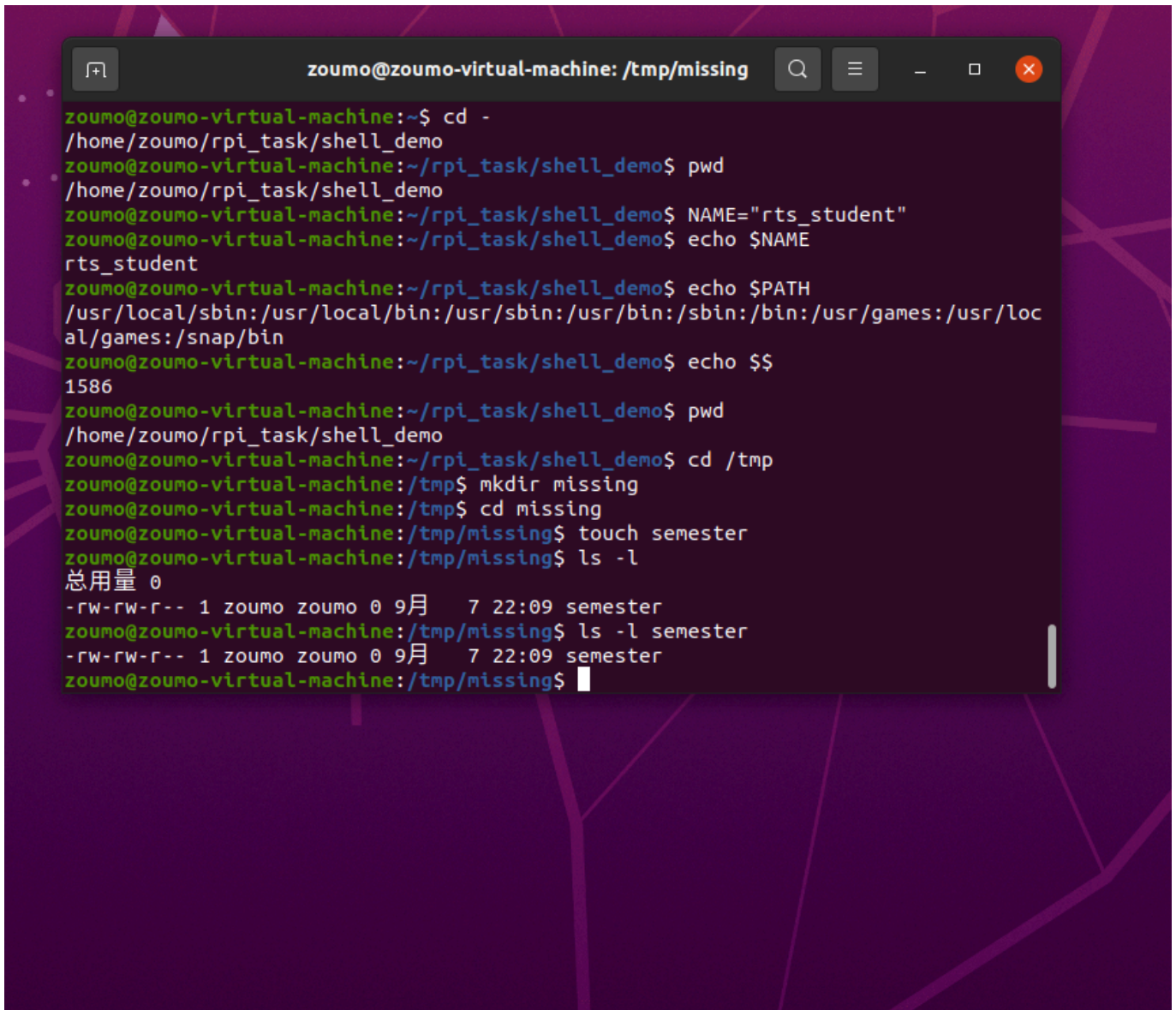
```
ls *.txt
```



```
zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task$ pwd
/home/zoumo/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ cd -
/home/zoumo
zoumo@zoumo-virtual-machine:~$ cd /home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cd -
/home/zoumo
zoumo@zoumo-virtual-machine:~$ pwd
/home/zoumo
zoumo@zoumo-virtual-machine:~$ cd -
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ NAME="rts_student"
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo $NAME
rts_student
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo $$
1586
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.10 目录操作练习

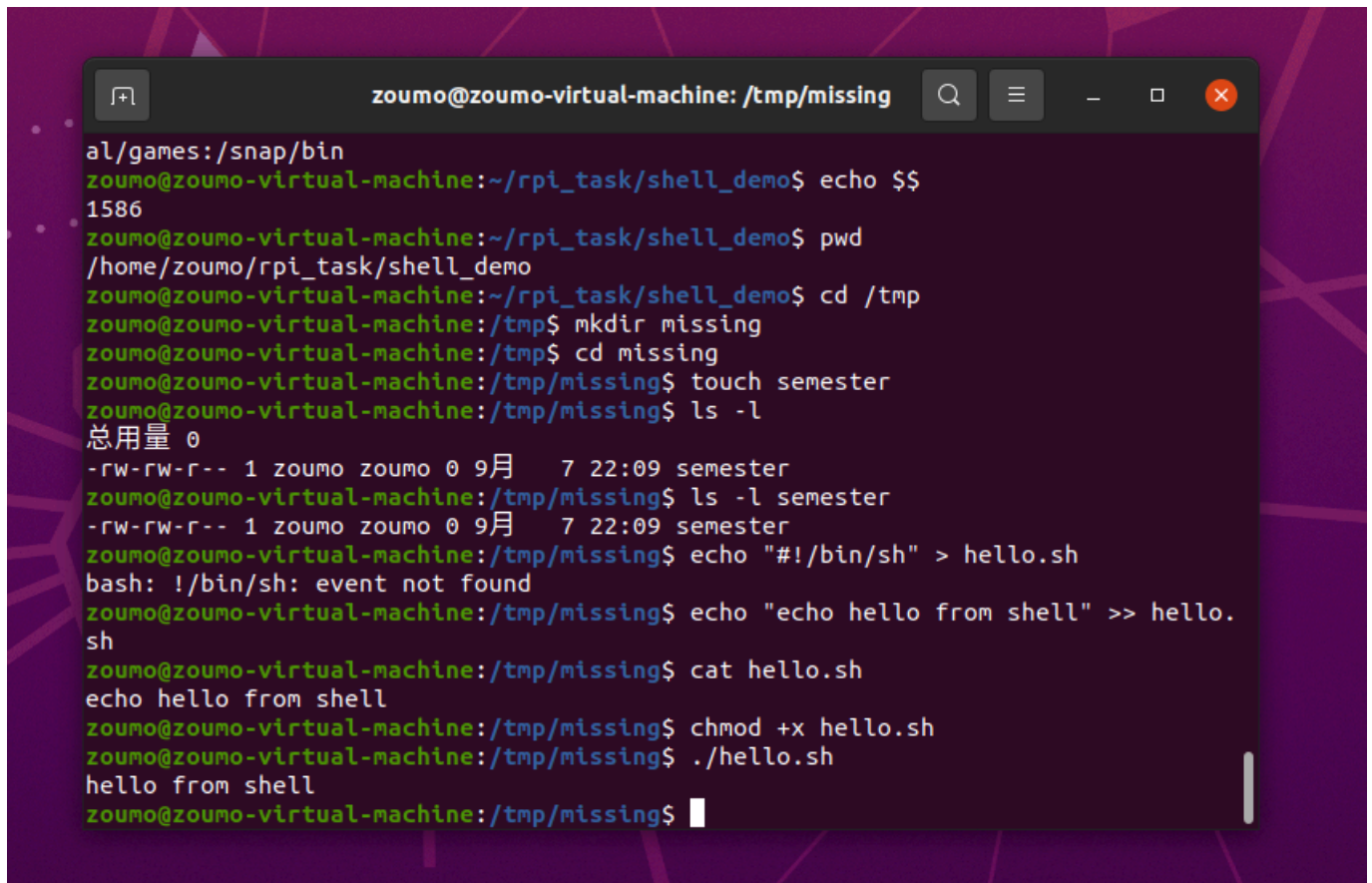
```
cd -
```

A terminal window titled 'zoumo@zoumo-virtual-machine: /tmp/missing' with standard window controls. The terminal shows a series of commands and their outputs. The user starts in the home directory, navigates to ~/rpi_task/shell_demo, sets the environment variable NAME to 'rts_student', and echoes it. Then they echo the \$PATH variable. Next, they echo the \$\$ variable, which shows the PID 1586. They then navigate to /tmp, create a directory named 'missing', and enter it. Inside /tmp/missing, they create a file named 'semester' using 'touch' and list the directory contents with 'ls -l'. The output shows a file named 'semester' with permissions -rw-rw-r-- owned by zoumo, created on September 7 at 22:09. Finally, they list the file specifically with 'ls -l semester' and get the same output. The prompt is currently at /tmp/missing\$.

```
zoumo@zoumo-virtual-machine:~$ cd -  
/home/zoumo/rpi_task/shell_demo  
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd  
/home/zoumo/rpi_task/shell_demo  
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ NAME="rts_student"  
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo $NAME  
rts_student  
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo $PATH  
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin  
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo $$  
1586  
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd  
/home/zoumo/rpi_task/shell_demo  
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cd /tmp  
zoumo@zoumo-virtual-machine:/tmp$ mkdir missing  
zoumo@zoumo-virtual-machine:/tmp$ cd missing  
zoumo@zoumo-virtual-machine:/tmp/missing$ touch semester  
zoumo@zoumo-virtual-machine:/tmp/missing$ ls -l  
总用量 0  
-rw-rw-r-- 1 zoumo zoumo 0 9月  7 22:09 semester  
zoumo@zoumo-virtual-machine:/tmp/missing$ ls -l semester  
-rw-rw-r-- 1 zoumo zoumo 0 9月  7 22:09 semester  
zoumo@zoumo-virtual-machine:/tmp/missing$
```

2.11 环境变量操作

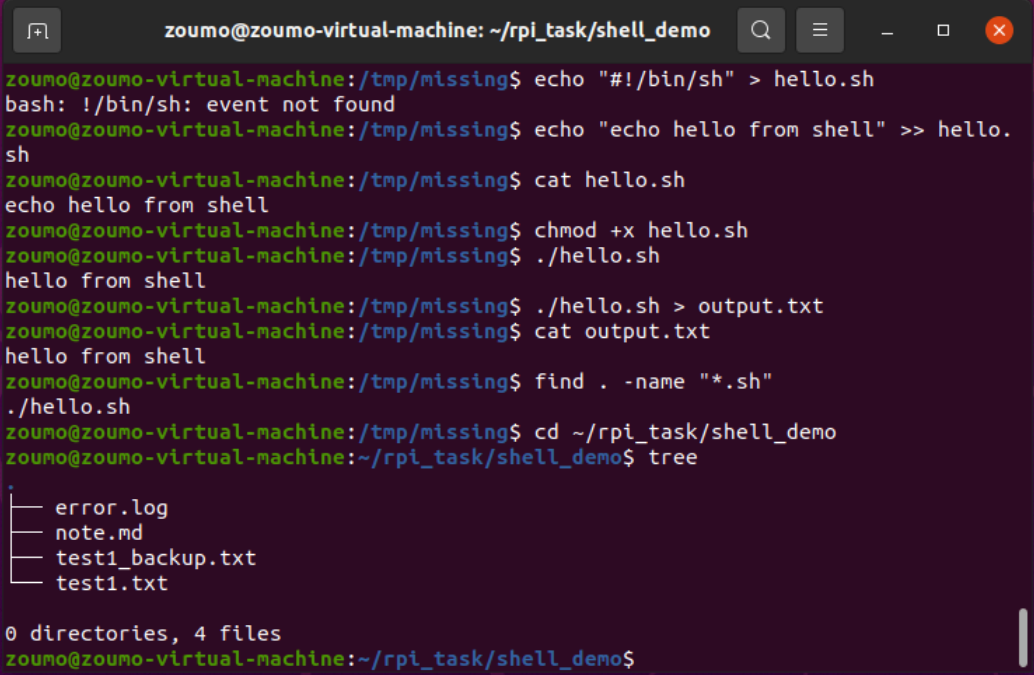
```
export TEST=123  
echo $TEST
```

A terminal window titled 'zoumo@zoumo-virtual-machine: /tmp/missing' with standard window controls. The terminal shows a series of commands and their outputs for setting up a shell. The user starts in a directory, runs 'echo \$\$' to get the PID (1586), then 'pwd' to see the current path, followed by 'cd /tmp'. They create a 'missing' directory, move into it, and create a file named 'semester' using 'touch'. Then they run 'ls -l' which shows the file with permissions and a date in Chinese. They then run 'ls -l semester' to show the file details. Next, they create a shell script 'hello.sh' with 'echo "#!/bin/sh" > hello.sh', but get an error 'bash: !/bin/sh: event not found'. They then use 'echo "echo hello from shell" >> hello.sh' to append the command. Finally, they run 'cat hello.sh' to see the script content, 'chmod +x hello.sh' to make it executable, and './hello.sh' to run it, which outputs 'hello from shell'.

```
al/games:/snap/bin
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ echo $$
1586
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ pwd
/home/zoumo/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cd /tmp
zoumo@zoumo-virtual-machine:/tmp$ mkdir missing
zoumo@zoumo-virtual-machine:/tmp$ cd missing
zoumo@zoumo-virtual-machine:/tmp/missing$ touch semester
zoumo@zoumo-virtual-machine:/tmp/missing$ ls -l
总用量 0
-rw-rw-r-- 1 zoumo zoumo 0 9月  7 22:09 semester
zoumo@zoumo-virtual-machine:/tmp/missing$ ls -l semester
-rw-rw-r-- 1 zoumo zoumo 0 9月  7 22:09 semester
zoumo@zoumo-virtual-machine:/tmp/missing$ echo "#!/bin/sh" > hello.sh
bash: !/bin/sh: event not found
zoumo@zoumo-virtual-machine:/tmp/missing$ echo "echo hello from shell" >> hello.
sh
zoumo@zoumo-virtual-machine:/tmp/missing$ cat hello.sh
echo hello from shell
zoumo@zoumo-virtual-machine:/tmp/missing$ chmod +x hello.sh
zoumo@zoumo-virtual-machine:/tmp/missing$ ./hello.sh
hello from shell
zoumo@zoumo-virtual-machine:/tmp/missing$
```

2.12 git 仓库初始化练习

```
git init
git config user.name "邹墨"
```

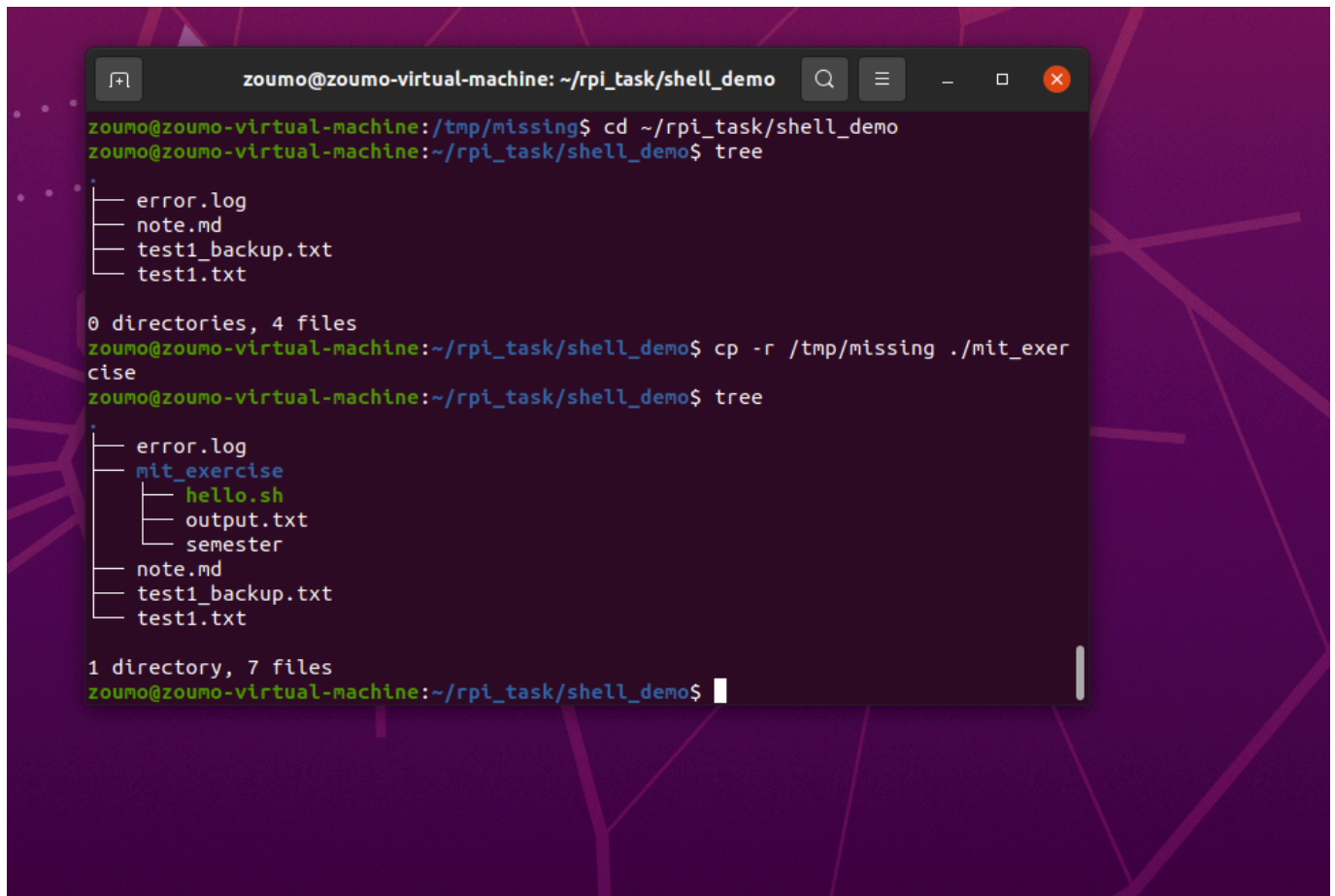


```
zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:/tmp/missing$ echo "#!/bin/sh" > hello.sh
bash: !/bin/sh: event not found
zoumo@zoumo-virtual-machine:/tmp/missing$ echo "echo hello from shell" >> hello.sh
zoumo@zoumo-virtual-machine:/tmp/missing$ cat hello.sh
echo hello from shell
zoumo@zoumo-virtual-machine:/tmp/missing$ chmod +x hello.sh
zoumo@zoumo-virtual-machine:/tmp/missing$ ./hello.sh
hello from shell
zoumo@zoumo-virtual-machine:/tmp/missing$ ./hello.sh > output.txt
zoumo@zoumo-virtual-machine:/tmp/missing$ cat output.txt
hello from shell
zoumo@zoumo-virtual-machine:/tmp/missing$ find . -name "*.sh"
./hello.sh
zoumo@zoumo-virtual-machine:/tmp/missing$ cd ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ tree
.
├── error.log
├── note.md
├── test1_backup.txt
└── test1.txt

0 directories, 4 files
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.13 git add 添加暂存

```
git add task1_report.md
```

A terminal window titled 'zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo'. The user navigates to ~/tmp/missing and runs 'tree', showing a directory with 4 files: error.log, note.md, test1_backup.txt, and test1.txt. Then, they run 'cp -r /tmp/missing ./mit_exercise' and run 'tree' again, showing the new directory structure with 7 files and 1 directory (mit_exercise).

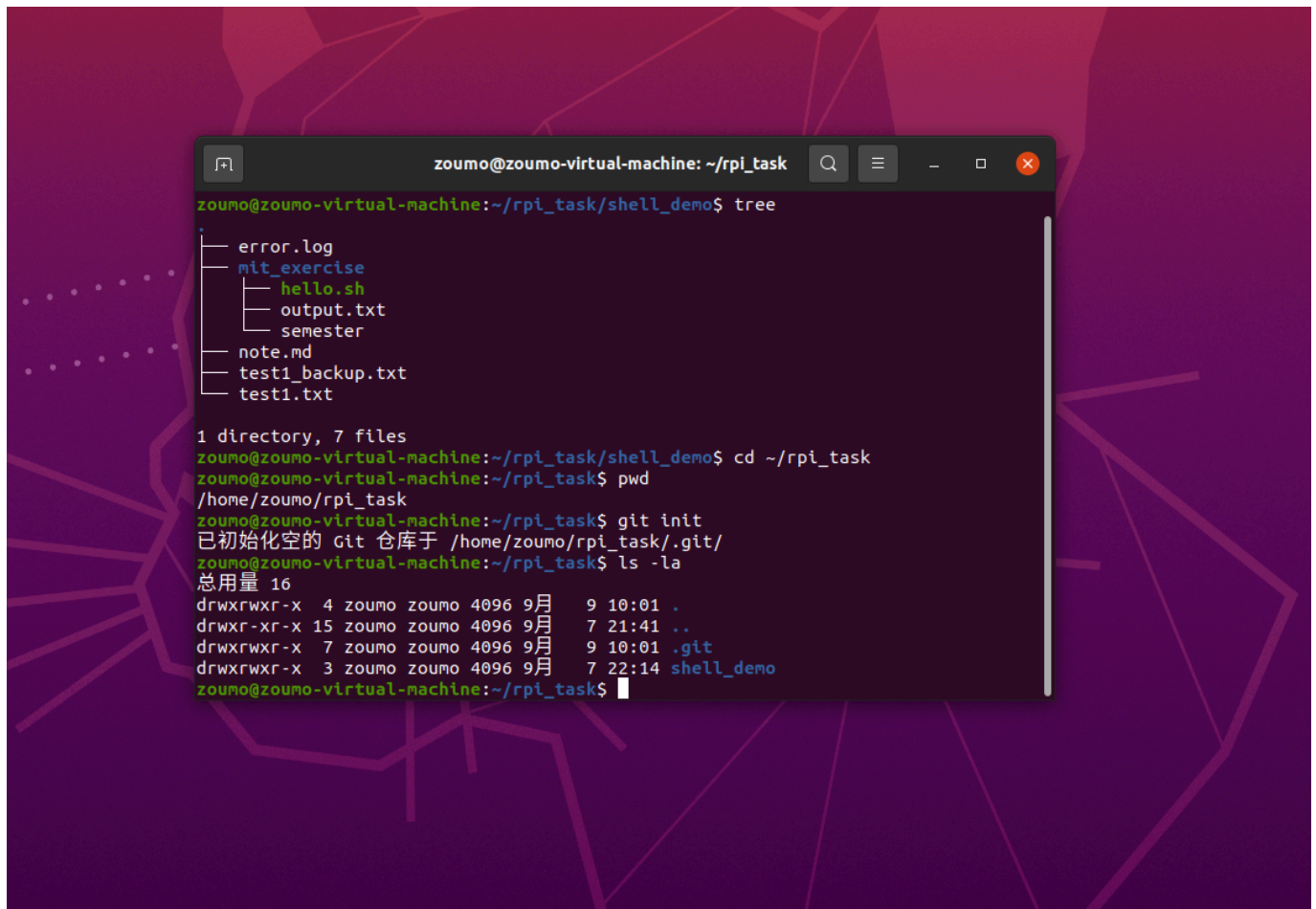
```
zoumo@zoumo-virtual-machine: ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:/tmp/missing$ cd ~/rpi_task/shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ tree
.
├── error.log
├── note.md
├── test1_backup.txt
└── test1.txt

0 directories, 4 files
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cp -r /tmp/missing ./mit_exercise
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ tree
.
├── error.log
├── mit_exercise
│   ├── hello.sh
│   ├── output.txt
│   └── semester
├── note.md
├── test1_backup.txt
└── test1.txt

1 directory, 7 files
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$
```

2.14 git commit 提交

```
git commit -m "add task1 report"
```

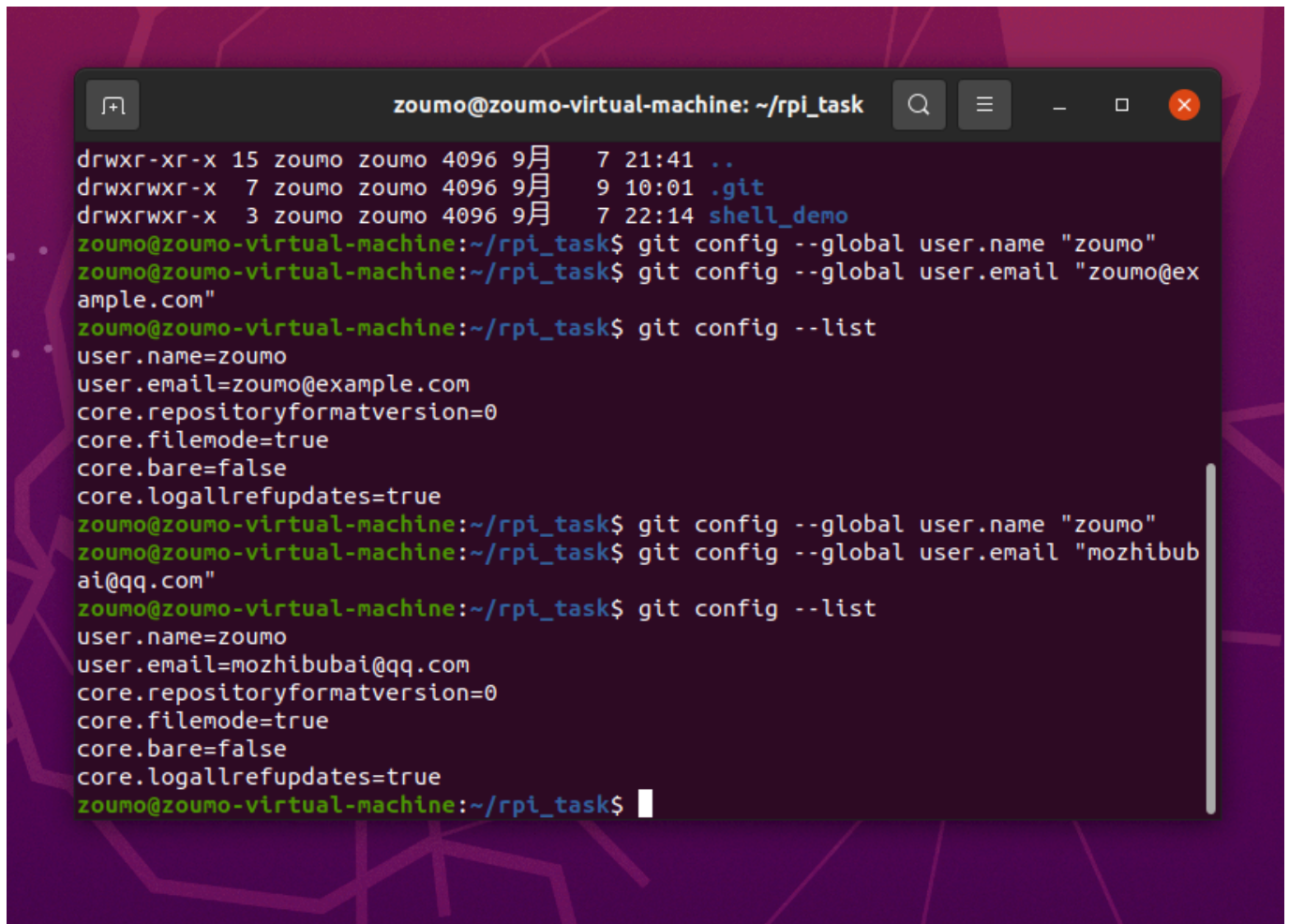
A terminal window titled 'zoumo@zoumo-virtual-machine: ~/rpi_task' is shown against a purple geometric background. The terminal displays the output of the 'tree' command in the '~/rpi_task/shell_demo' directory, showing files like 'error.log', 'mit_exercise', 'hello.sh', 'output.txt', 'semester', 'note.md', 'test1_backup.txt', and 'test1.txt'. It then shows the user navigating to '~/rpi_task', running 'pwd' to confirm the path, and 'git init' to initialize a git repository. The final command is 'ls -la', which shows the newly created '.git' directory and the 'shell_demo' subdirectory.

```
zoumo@zoumo-virtual-machine: ~/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ tree
.
├── error.log
├── mit_exercise
│   ├── hello.sh
│   ├── output.txt
│   └── semester
├── note.md
├── test1_backup.txt
└── test1.txt

1 directory, 7 files
zoumo@zoumo-virtual-machine:~/rpi_task/shell_demo$ cd ~/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ pwd
/home/zoumo/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ git init
已初始化空的 git 仓库于 /home/zoumo/rpi_task/.git/
zoumo@zoumo-virtual-machine:~/rpi_task$ ls -la
总用量 16
drwxrwxr-x  4 zoumo zoumo 4096 9月  9 10:01 .
drwxr-xr-x 15 zoumo zoumo 4096 9月  7 21:41 ..
drwxrwxr-x  7 zoumo zoumo 4096 9月  9 10:01 .git
drwxrwxr-x  3 zoumo zoumo 4096 9月  7 22:14 shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task$
```

2.15 git 分支创建

```
git branch dev
git checkout dev
```

A terminal window titled 'zoumo@zoumo-virtual-machine: ~/rpi_task' with standard window controls. It shows a directory listing of files: '..', '.git', and 'shell_demo'. Then, it shows a series of git configuration commands: setting user.name to 'zoumo', user.email to 'zoumo@example.com', listing the current config, then changing user.email to 'mozhibubai@qq.com', and listing the config again. The prompt is at the end of the last line.

```
zoumo@zoumo-virtual-machine: ~/rpi_task
drwxr-xr-x 15 zoumo zoumo 4096 9月 7 21:41 ..
drwxrwxr-x  7 zoumo zoumo 4096 9月 9 10:01 .git
drwxrwxr-x  3 zoumo zoumo 4096 9月 7 22:14 shell_demo
zoumo@zoumo-virtual-machine:~/rpi_task$ git config --global user.name "zoumo"
zoumo@zoumo-virtual-machine:~/rpi_task$ git config --global user.email "zoumo@example.com"
zoumo@zoumo-virtual-machine:~/rpi_task$ git config --list
user.name=zoumo
user.email=zoumo@example.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
zoumo@zoumo-virtual-machine:~/rpi_task$ git config --global user.name "zoumo"
zoumo@zoumo-virtual-machine:~/rpi_task$ git config --global user.email "mozhibubai@qq.com"
zoumo@zoumo-virtual-machine:~/rpi_task$ git config --list
user.name=zoumo
user.email=mozhibubai@qq.com
core.repositoryformatversion=0
core.filemode=true
core.bare=false
core.logallrefupdates=true
zoumo@zoumo-virtual-machine:~/rpi_task$
```

2.16 git 分支合并

```
git checkout master
git merge dev
```

```
zoumo@zoumo-virtual-machine: ~/rpi_task

新文件:      shell_demo/mit_exercise/output.txt
新文件:      shell_demo/mit_exercise/semester
新文件:      shell_demo/note.md
新文件:      shell_demo/test1.txt
新文件:      shell_demo/test1_backup.txt

zoumo@zoumo-virtual-machine:~/rpi_task$ git commit -m "feat: add task1 shell demo files"
[master (根提交) 193e2ba] feat: add task1 shell demo files
7 files changed, 7 insertions(+)
create mode 100644 shell_demo/error.log
create mode 100755 shell_demo/mit_exercise/hello.sh
create mode 100644 shell_demo/mit_exercise/output.txt
create mode 100644 shell_demo/mit_exercise/semester
create mode 100644 shell_demo/note.md
create mode 100644 shell_demo/test1.txt
create mode 100644 shell_demo/test1_backup.txt
zoumo@zoumo-virtual-machine:~/rpi_task$ git log
commit 193e2baef8ff10e135a7c7e2c644e56fa214995e (HEAD -> master)
Author: zoumo <mozhibubai@qq.com>
Date:   Wed Sep 9 10:09:48 2026 +0800

    feat: add task1 shell demo files
zoumo@zoumo-virtual-machine:~/rpi_task$ S
```

2.17 git 撤销修改

```
git checkout -- test.txt
```

```
zoumo@zoumo-virtual-machine: ~/rpi_task
zoumo@zoumo-virtual-machine:~/rpi_task$ git status
位于分支 master

尚无提交

未跟踪的文件:
  (使用 "git add <文件>..." 以包含要提交的内容)
    shell_demo/

提交为空，但是存在尚未跟踪的文件（使用 "git add" 建立跟踪）
zoumo@zoumo-virtual-machine:~/rpi_task$ git add shell_demo/
zoumo@zoumo-virtual-machine:~/rpi_task$ git status
位于分支 master

尚无提交

要提交的变更:
  (使用 "git rm --cached <文件>..." 以取消暂存)
    新文件:   shell_demo/error.log
    新文件:   shell_demo/mit_exercise/hello.sh
    新文件:   shell_demo/mit_exercise/output.txt
    新文件:   shell_demo/mit_exercise/semester
    新文件:   shell_demo/note.md
    新文件:   shell_demo/test1.txt
```

2.18 git log 查看提交记录

```
git log
```

```
zoumo@zoumo-virtual-machine: ~/rpi_task

zoumo@zoumo-virtual-machine:~/rpi_task$ git log
commit 193e2baef8ff10e135a7c7e2c644e56fa214995e (HEAD -> master)
Author: zoumo <mozhibubai@qq.com>
Date:   Wed Sep 9 10:09:48 2026 +0800

    feat: add task1 shell demo files
zoumo@zoumo-virtual-machine:~/rpi_task$ echo "# Task1 学习笔记" >> shell_demo/note.md
zoumo@zoumo-virtual-machine:~/rpi_task$ git status
位于分支 master
尚未暂存以备提交的变更:
  (使用 "git add <文件>..." 更新要提交的内容)
  (使用 "git restore <文件>..." 丢弃工作区的改动)
    修改:   shell_demo/note.md

修改尚未加入提交 (使用 "git add" 和/或 "git commit -a")
zoumo@zoumo-virtual-machine:~/rpi_task$ git diff
diff --git a/shell_demo/note.md b/shell_demo/note.md
index e69de29..a3e8925 100644
--- a/shell_demo/note.md
+++ b/shell_demo/note.md
@@ -0,0 +1 @@
+# Task1 学习笔记
zoumo@zoumo-virtual-machine:~/rpi_task$
```

2.19 整体实验环境

```
zoumo@zoumo-virtual-machine: ~/rpi_task

diff --git a/shell_demo/note.md b/shell_demo/note.md
index e69de29..a3e8925 100644
--- a/shell_demo/note.md
+++ b/shell_demo/note.md
@@ -0,0 +1 @@
+# Task1 学习笔记
zoumo@zoumo-virtual-machine:~/rpi_task$ git add shell_demo/note.md
zoumo@zoumo-virtual-machine:~/rpi_task$ git commit -m "docs: add title to note.md"
[master 01a1b42] docs: add title to note.md
 1 file changed, 1 insertion(+)
zoumo@zoumo-virtual-machine:~/rpi_task$ git log
commit 01a1b427356e1140ae4fea7394f4b5d04e51b686 (HEAD -> master)
Author: zoumo <mozhibubai@qq.com>
Date:   Wed Sep 9 10:13:20 2026 +0800

    docs: add title to note.md

commit 193e2baef8ff10e135a7c7e2c644e56fa214995e
Author: zoumo <mozhibubai@qq.com>
Date:   Wed Sep 9 10:09:48 2026 +0800

    feat: add task1 shell demo files
zoumo@zoumo-virtual-machine:~/rpi_task$
```

三、学习总结 本次实验在 Ubuntu 虚拟机中学习 Linux Shell 基础命令，掌握了目录跳转、文件创建复制删除、管道与文本筛选等基础操作。同时学习 Git 版本控制工具，学会初始化仓库、提交文件、创建合并分支、撤销

修改。熟悉了命令行的工作逻辑，也学会使用 Markdown 编写实验报告，理解相对路径引用图片，为后续 RTS 视觉算法组任务打下基础。